RIDEBOUND

Final source, logic, benchmark and evidence review

Work packages WP1–WP10 · post-confirmatory closure

> Outcome
> Repository assurance gates pass, but the scientific results stay negative: WP9 fails the service gate at both capacity strata, and WP10 fails closed on an explicit RidePy position capability.

Review date: 23 August 2026

Evidence authority: repository docs/18, docs/19, ADR-048/051/052 and immutable external receipts

Claim class: finite-panel empirical result + mechanical correctness/reproducibility. No population, SLA, satisfaction or novelty claim.

1. Executive verdict

No unresolved correctness or evidence-integrity blocker was found in the reviewed RideBound repository. This is not a formal proof: it is the result of a full-tree machine pass, manual review of outcome-bearing paths, differential/mutation tests, exact artifact binding and actual simulator execution.

WP9 · confirmatory result
FAIL. Panel A (8 vehicles): 1735 → 1581 completed, −154 = −7.1296 pp. Panel B (4 vehicles): 966 → 860, −106 = −4.9074 pp. Both exceed the preregistered −1.00 pp margin.

WP10 · cross-system result
NEGATIVE CAPABILITY. Canonical RidePy execution passes. The representative subset fails closed because nodeOnly cannot represent concurrent mid-edge vehicle progress. Layer 3 is not established.

Engineering closure
855/855 .NET, 95/95 FleetPy and 23/23 RidePy tests pass; Release and format are clean; no known vulnerable NuGet package. ADR-052 removes exact redundant cache-key allocation without changing any search counter.

What this report does not claim

It does not convert near-zero revision burden into user benefit, infer a population effect from five travel days, rescue failed cells, claim production latency, or claim dynamic insertion / ETA stability / least commitment / satisfaction as novel.

2. Review scope and method

The pre-report inventory opened 1,226 reviewable files (302,565,761 bytes; 145,082 text lines). The benchmark tool and later patches were read directly. The final scanner covered 1,232 reviewable files, 1,221 text files and 146,534 text lines.

Review layer	What was checked	Why it matters
Whole tree	UTF-8, JSON, Python syntax, Markdown links/fences, imports, stubs, nondeterminism, float and dependency scans	Finds defects beyond selected unit-test paths
Manual high-risk read	Protocol/hash/retry; reducer/physical; ledger/budget/locks; candidate/solver; bundles/oracles; FleetPy/RidePy; WP9/WP10 analyzers	Targets outcome-bearing and fail-open boundaries
Independent evidence	Exact-small oracles, process oracle, mutation tests, frozen receipts, terminal inventories	Avoids trusting a producer's self-reported success
Actual systems	Pinned FleetPy 1.0.2, RidePy 2.10.1 container and read-only BeGo baseline	Separates adapter mocks from native lifecycle behavior
Claim audit	Finite-panel unit, service gate, locked/earned burden, prior-art boundary	Prevents a mechanically correct artifact from supporting an invalid claim

Architecture result

Domain and Application contain no EF Core, ASP.NET, map-provider, OR-Tools, FleetPy, RidePy, OptiGo or Npgsql dependency. OR-Tools remains isolated in its solver adapter. Python adapters perform mapping and lifecycle orchestration and call the same versioned Runner; static scan found no alternate solver, budget or hard-lock implementation.

3. Work-package verdicts

WP	Boundary revalidated	Current verdict
1	Strict protocol, canonical JSON/hash, retry/session	PASS — no ambiguity or replay advance found
2	Immutable online state, route/frozen prefix, physical validator, B1	PASS — conservation and no-reassignment enforced
3	Promises, three-way delta, 10D ledger/budget, locks, certificates	PASS — monotone/no-refund; exact state binding
4	Bounded candidates, portfolio, CP-SAT/fallback/evidence	PASS within declared bounds; no global-optimality overclaim
5	Independent BeGo durable adapter and recorded DB/process evidence	PASS for mechanical integration; no effectiveness upgrade
6	Dataset/plan/process/metric/oracle/bundle verifier	PASS — verifier recomputes outcome-bearing fields
7	FleetPy mapping/clock/plan/Runner client	PASS — actual pinned Layer 2 mechanics
8	Experimental unit, pairing, endpoint, panel, prereg/freeze	PASS for finite-panel design; solver seed not a replicate
9	H6 panels, burden decomposition, robustness, reproducibility	COMPLETE — negative confirmatory result
10	RidePy source/image/native lifecycle/subset/failure retention	COMPLETE — negative capability result

A completed work package means its declared gate was evaluated and its evidence retained. It does not mean the treatment won. WP9 and WP10 are correctly complete with negative results.

4. WP9 confirmatory result

Service-rate delta versus preregistered −1.00 pp margin

Panel	Arrivals / arm	B1 completed	C1 completed	Delta	Gate
A · 8 vehicles	2,160	1,735	1,581	−154 · −7.1296 pp	FAIL
B · 4 vehicles	2,160	966	860	−106 · −4.9074 pp	FAIL

Why the burden gate does not rescue service

In Panel A, C1 burden is 0.17% of B1 and exactly zero in 12/20 cells. Pickup-ETA burden is definitional under the lock. Much of the remaining reduction occurs because C1 declines work rather than serving it with fewer revisions. Robustness attributes approximately half the service loss to lock/ranking and half to the 30-second budget; C2 recovers almost nothing.

All 20 Panel A cells are negative. Solver seed 19 minus seed 7 is exactly zero on completed service, burden and disruptive count in both arms; seeds do not create independent units. The conclusion is conditional on the fixed panels and five travel realizations, with achieved precision about 1.40 pp against the 1.00 pp margin.

5. WP10 RidePy result

Canonical gate · PASS
Both B1 and C1 complete 5/5 requests, reconcile 5 native pickups + 5 native drops and emit 22 Runner decisions. The published Runner tree remains byte-identical.

Representative subset · FAIL CLOSED
22 arm jobs pass. B1 travel-update-stress-r3 fails at epoch 17; its paired C1 arm is not run. Native pickup occurs at 116,000 ms while the last nodeOnly Runner ETA remains 178,000 ms: a 62-second observability gap.

Valid-pair descriptive summary

Stratum	B1	C1	Delta service
Uncongested	—	—	0.00 pp
Insertion	—	—	−7.14 pp
Travel stress	—	—	−16.67 pp
All 11 valid pairs	54 / 62	49 / 62	−8.06 pp

These 11 pairs are not the planned complete subset estimand. No missing-job denominator is laundered, no confidence interval is reported, and the result is not pooled with H6. The named limitation is RBWP10_NODEONLY_CONCURRENT_MIDEDGE_UNSUPPORTED; inferring progress from clock would invent simulator state.

Runtime recoverability

The exact executed image is archived at 695,427,072 bytes with SHA-256 4783c541c256d1551677684eb5182cc43a8845d6bb3c5dc34778aadc9fc9a872 and reloads to image ID sha256:5468b9cba13c…e573. This proves exact restore, not future byte-identical rebuilding from evolving apt/pip indexes.

6. Full-PDF research audit

The optimization was selected after extracting and reading every page of the retained PDFs, not from titles or abstracts. Hashes below bind the exact files. Full untruncated hashes and extraction hashes are recorded in docs/21.

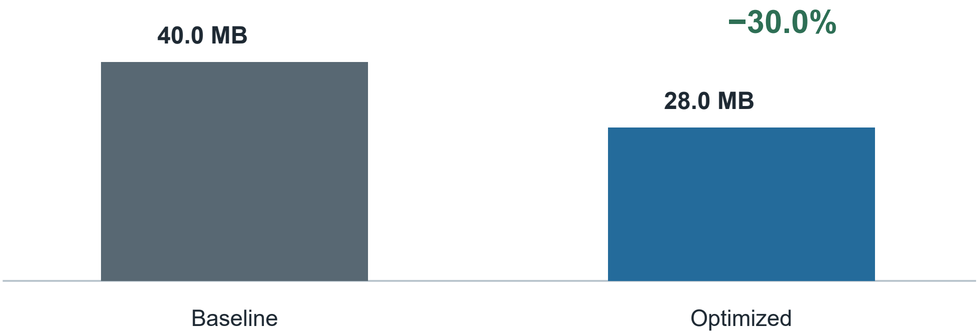
Full PDF	Pages	SHA-256	Applied / rejected conclusion
Alonso-Mora et al. 2017 · main	6	edbb62156e36479b…64c12ea	Keep request–trip–vehicle separation and bounded feasible sets; do not import reassignment
Alonso-Mora et al. 2017 · supplement	32	0d7e37aba541035b…760ddd5	Reuse requires exact state identity
Gschwind & Drexl 2019	39	16b82b489c6ae925…ad18e61	Exact temporal preprocessing is relevant; full constant-time test not transplanted
Simonetto et al. 2019	30	9f5e31a8d69a63b1…ae3868	Sparse/batched LAP changes the pool; no pre-confirmatory import
Engelhardt et al. 2020	11	5b3d20b26e701da7…fd01ed	Direction/distance/random filters trade quality for time; rejected without loss bound
Zalesak et al. 2025	23	744193567e9033de…8ba086	Route stability and assignment mechanisms are prior art and distinct
Schulz & Pfeiffer 2026	46	9a4d4997cdecc724…5c8d7	Preprocess/reuse needs explicit invalidation; paper horizons are not defaults

Research boundary
Literature is evidence for a mechanism and its assumptions, not a source of RideBound defaults. No random/direction/sparse prune, reassignment, new batching interval, promise horizon, service margin or novelty claim was imported.

7. ADR-052 exact-reuse optimization

ForwardSlackCacheKey allocated a textual position fingerprint on every lookup even though the same key already held the immutable VehicleState reference. Terminal evaluation also constructed a second key only to verify the prefetched lookup created by that exact node. The patch removes the redundant string and compares the existing key against exact run, vehicle, route, time, travel and allowance inputs.

Cache-key allocation per 250,000 constructions



Route stops	Baseline μs / 250k	Optimized	Time	Allocation
4	48,170	43,955	−8.8%	40,000,072 → 28,000,072 · −30%
8	58,469	54,894	−6.1%	40,000,072 → 28,000,072 · −30%
16	80,861	77,082	−4.7%	40,000,072 → 28,000,072 · −30%

End-to-end generator

Allocation falls consistently by 0.79–1.30%. Timing is mixed (−4.9%, +2.8%, −1.9%), so this report makes no generator speed claim. All six work-profile counters remain identical in every baseline and optimized process; the full physical validator still runs.

8. Verification and review findings

Gate	Observed result
RideBound required suite	855 pass · 0 fail · 0 skip
Release /warnaserror	0 warnings · 0 errors
Format + diff	verify-only format PASS; git diff --check PASS
NuGet vulnerability	0 known vulnerable direct/transitive package in all solution projects
FleetPy pinned suite	95/95 · 0 skip
RidePy pinned-container suite	23/23
BeGo read-only current baseline	backend 149 pass + 5 explicit opt-in skip; frontend 9/9
Final tree scan	1,232 files; 1,221 text; 146,534 lines; 0 UTF-8/JSON/Python/Markdown error
Architecture/static	0 forbidden core dependency; 0 decision RNG/wall clock/float; 0 Python decision reimplementation
WP10 analyzer	exact terminal inventory + freeze/full Runner/seed; 7 mutation classes
Docker archive/load	SHA-256 exact; reload returns same image ID

Defects found by review, not test-count worship

The strengthened WP10 analyzer now rejects an omitted valid pair, any extra/unplanned arm or failure artifact, seed drift and Runner-receipt drift. Format review corrected line-ending/import drift. The image archive closes exact restore risk. None of these changes modifies WP9 H6 or the WP10 terminal failure.

9. Residual risks and conclusion

Residual risk	Current control / honest limit
Finite-panel inference	Report only the 20-cell/five-travel-realization result; no population CI or universal claim
Bounded candidate generation	Loss diagnostics are explicit, but no general solution-quality bound
RidePy mid-edge observability	Named fail-closed capability; do not infer hidden state
Container rebuild	Exact image archive restores; Dockerfile rebuild is not claimed byte-identical
Dirty worktree ownership	Preserved user artifacts/config/corpora/__pycache__; no cleanup/reset
Review strength	Whole-tree + manual + mutation + native simulators is strong assurance, not formal verification

Final conclusion
WP1–WP10 are implementation/evidence complete at their declared gates. The repository baseline is green and the final review found no unresolved blocker. The treatment nevertheless fails the WP9 service criterion, and RidePy does not establish Layer 3. These negative results are the correct terminal findings—not defects to hide or tune away.

Next decision

Do not silently proceed to a new effectiveness experiment. WP11 Product UX or WP12 manuscript/release requires a new refinement decision that preserves H6, exposes limitations and treats intermediate commitment levels as exploratory rather than retroactive rescue.

Appendix A. Evidence identities

Evidence	Identity / location
WP9 H6 freeze	84f6eff31addbdd12349a19201d79c872fbd05aaf5e0aa45dd73aee6d5c3dee2
WP10 source/runtime receipt	2b43106207b142e7ccde39482f73d551678881b869f0954cdc638ec9e7840775
WP10 subset freeze v3	18a74fa34f94a35ff92fbdc4ea2611e982527682760c52432419e0feef206672
WP10 strengthened analysis v2	be3e90771ae0216e891a8284b5457b1f635db214aa52681ce8d468b4007dcca3
WP10 failure transcript	0ee5e3ec…a85 · retained under E:\RideBoundData\wp10\subset-results-v3
Exact RidePy image archive	4783c541c256d1551677684eb5182cc43a8845d6bb3c5dc34778aadc9fc9a872
Optimization raw JSON	six SHA-256 values in docs/benchmarking/post-wp10-exact-reuse-optimization-2026-08-23.md
Repository review	docs/reviews/wp1-wp10-final
Live status / traceability	docs/18-status-and-decision-log.md · docs/19-requirement-traceability.md

Interpretation guard

Hashes establish exact artifact identity, not correctness by themselves. Correctness support comes from independent recomputation, mutation/differential tests, explicit failure retention, source review and actual simulator execution. All evidence should be read together with the claim boundary in docs/03.

Appendix B. Full-PDF provenance

Original files are retained outside the repository at E:\RideBoundData\research\pdf-20260820. Every page was extracted and inspected; the hashes below are complete rather than display abbreviations.

Retained PDF	Pages	Exact SHA-256
Alonso-Mora et al. 2017 · main	6	edbb62156e36479b742a1a7381e59206 73a4b6a3130bba39aed74cb8364c12ea
Alonso-Mora et al. 2017 · supplement	32	0d7e37aba541035bdbc60da0eb35e81a 63859eb96850a28d9a8fba116760ddd5
Gschwind & Drexl 2019	39	16b82b489c6ae925581bebd00223d4aa8 bce7541f1b561b2bd2320529ad18e61
Simonetto et al. 2019	30	9f5e31a8d69a63b1286fbe55577d07a 8449585356aa6802df9fb734d75ae3868
Engelhardt et al. 2020	11	5b3d20b26e701da7837a149eb2953e1a 828d0bf4fab780dbe5a50eb6defd01ed
Zalesak et al. 2025	23	744193567e9033de631bc636045302399 55d70575ddb8614b7d75fcf078ba086
Schulz & Pfeiffer 2026	46	9a4d4997cdecc7242521ff733f8d474b 6b57dc1856ce261734b7923b25a8c8d7

The paper evidence constrains what may be reused; it does not turn a cited mechanism into RideBound novelty or an empirical result on the RideBound panels.

End of report